



Vidit Bohra
Mechanical Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 13/09/2006

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	8.58 Core: 128 Total: 128
Intermediate	HSC	Pace Junior Science College, Dadar	2024	86.67%
Matriculation	CISCE	Campion School	2022	96.20%

Pursuing a Minor in Artificial Intelligence and Data Science

RESEARCH EXPERIENCE

Human-Object Interaction Retargeting for Robot Manipulation | Utokyo (Jun '26 - Jul '26)

Guide: Prof. Gentiane Venture | GV Lab

- Built an end-to-end pipeline converting monocular RGB video into robot motion using HMR2 (4D-Humans) pose estimation, SMPL joint extraction, and IK retargeting (Holosoma), deployed on a Pepper humanoid and UR5 arm
- Leveraged the **interaction-mesh based retargeting** formulation to preserve fine-grained **human-object spatial relationships**, ensuring retargeted motion retains complete **manipulation semantics** instead of copying joint angles

TECHNICAL EXPERIENCE

Mars Rover Team | IIT Bombay

(Oct '24 - Ongoing)

Guide: Prof. P.J. Guruprasad, Dept. of Aerospace Engineering, IIT Bombay

- Part of a team of **30+** students from various disciplines working together to make a **semi-autonomous rover** capable of traversing difficult terrains and performing complex tasks in competitions like **URC** and **IRC**.
- Ranked **2nd** in the **European Rover Challenge** out of **24** teams spanning **15** countries, including **100+** students.
- Secured **1st place** in the **Astrobiology Mission** at the **International Rover Challenge (IRC)**
- Ranked **9th** in the **IRC**, organized by SPROS, out of **35+** teams from various institutions worldwide.

Senior Design Engineer | Robotic Arm Subsystem | Mars Rover Team

(Apr '25 - Apr '26)

Guide: Prof. P.J. Guruprasad, Dept. of Aerospace Engineering, IIT Bombay

- Designed a **5-DOF arm** with various key changes and implementations to improve performance and control
- Enforced mechanisms including a custom **2 stage 81:1 cycloidal drive** and **explicit steering** for improved control
- Redesigned a **2-DOF differential wrist** into a **35% lighter**, 3D-printed version that's easier to manufacture
- Applied an **optical disk drive** inspired rail design to eliminate binding from tolerances in a scaled-up **linear base**
- Integrated a **worm gear drive** to prevent **back drive** and **maintain position** of the wrist in event of a power loss

KEY PROJECTS

Robo Sapient - Bipedal Robot

(May '25 - Ongoing)

ITSP (Institute Technical Summer Project)

- Conducted a literature review on various existing **bipedal robots** and **papers**, to establish **baseline requirements**
- Designed and modeled a 6-DOF biped robot using lightweight **carbon fiber tubes** to optimize weight and efficiency, and consisting of **limit sensors** and a **rack and pinion assembly** to change the location of the COM reliably
- Researched RL algorithms for locomotion and trained a **PPO policy** in Isaac Lab using a **multi-term reward function** including **velocity tracking**, **feet air time**, and **joint limit penalties**, achieving a stable walking gait
- Fabricated the biped using **FDM 3D printing** and successfully integrated **inverse kinematics (IK)** and **PID control loops** for real-time balance and stability, using an **Arduino Mega**, **MPU6050** and various sensors
- Implemented a **data-driven controller** in Simulink by fitting a **state-space model** from Simscape (**85% fit**)
- Designed an **LQR controller** on a **Linear Inverted Pendulum Model** to achieve a stable walking gait

Holo-Battalion

(Aug '25 - Ongoing)

E-Yantra Robotics Competition

- Designing a swarm of 3 bots, each equipped with a single **articulated arm** and **omnidirectional holonomic drive**
- Implemented a **ROS2 server-client** framework integrating ArUco marker vision with homography for pose estimation
- Prototyped three bots with a **custom etched PCB** for the ESP32 MCU and a **linear gripper assembly**
- Orchestrated a **multi-agent warehouse fleet** of three robots by implementing a **centralized task allocation algorithm** to optimize assignment based on Euclidean proximity, minimizing total travel time

Estimation Of Lie Groups

(Aug '25 - Oct'25)

Course Project | Estimation of Lie Groups SC651 | Prof. Ravi Banavar

- Implemented the **Kalman Filter**, **TRIAD** and **QUEST** on an IMU to acquire a stable estimates of the system
- Utilized the **Kalman Filter** to fuse sensor data from the **gyroscope**, **accelerometer**, and **magnetometer**
- Implemented the **Mahony filter** and systematically compared its stability and performance against the **TRIAD** and **QUEST** methods, achieving a **40% decrease** in the standard deviation of the output

OTHER PROJECTS

Wildlife Detection via Multi-Feature Engineering (Aug'25 - Oct '25)

Course Project | Programming for Data Science DS203 | Prof. Vinay Kulkarni

- Built an **XGBoost** pipeline to classify wildlife presence, improving accuracy **60%** over baseline logistic regression
- Built a high-dimensional feature set combining **HOG, LBP, GLCM, and ORB** descriptors and applied **SelectKBest** pruning to retain the top 150 most predictive variables, boosting accuracy by **40%** over a raw pixel-only baseline

Stress Concentration Analysis via Photoelasticity (Jan'26 - Apr'26)

Course Project | Solid Mechanics Lab ME218 | Prof. Sripriya Ramamoorthy

- Performed **photoelasticity** testing using a polariscope setup to compute **stress concentration factors (SCF)**
- Fabricated **TPU-molded epoxy** specimens with square, filleted-square, and elliptical hole geometries
- Validated results against **ANSYS FEA** simulations, showing strong agreement in stress concentration locations
- Filletting the square hole's sharp corners cut SCF by **21%**; **elliptical geometry** reduced it further by **33%**

Support-Free 3D Printing via Conical Slicing (Jan'26 - Apr'26)

Course Project | ME230 | Prof. Ramesh Singh

- Implemented **conical non-planar slicing** on an unmodified 3-axis FDM printer, fabricating a **90° overhang** support-free via **STL pre-warping**, planar slicing and finally **G-code back-transformation** in Python
- Achieved **25% material** and **20% build-time reduction** over conventional supported prints, eliminating post-processing

Drone Remote Controller (Jul'24 - Nov'24)

Course Project | Makerspace MS101 | Prof. Joseph John

- Fabricated a **joystick remote-control system** enabling flight control having hover capability for a drone prototype
- Developed hardware design by drafting schematics and routing **PCB in EasyEDA** for a controller powered by **ESP32**, equipped with ARM switches and OLED display and implemented **control-axis calibration** procedures

POSITIONS OF RESPONSIBILITY

Subsystem Lead | Robotic Arm & LDT | Mars Rover Team (Apr '26 - Present)

Interdisciplinary team of **25+ members** designing and building robotic subsystems for planetary rover competitions

- Mentored the design and construction of a **quadrupedal robot** and a **hexapod robot**, guiding **mechanical design**, actuator selection, and control integration from concept to working prototype
- Leading a team of **15** undergraduate students in the design and development of a **multi-functional robotic arm**
- Designed the robotic arm to ensure **minimal inertia** and **ease of control** by utilizing a combination of belt drives, BLDC motors and **2-stage planetary and cycloidal gearbox** to achieve precise motion

Convener | Krittika | Astronomy Club, IIT Bombay (May '25 - Mar '26)

Interdisciplinary team of **10+ members** making astronomy accessible through research and engagement initiatives

- Designed a custom **2-DOF stand** for the **EdgeHD11 telescope** installed in the newly inaugurated observatory
- Led a learners space session on **Coordinate Systems & Time** and led a talk on **Careers in Astronomy**
- Assisted in organizing an exhibition for **National Space Day** engaging over **100+ students** and **10k+ residents**
- Conducted a workshop for **300+ students**, making a constellation projector and sundial with Gnaanu organization

TECHNICAL SKILLS

Software	SolidWorks, ANSYS, MATLAB, MSC Adams, Fusion360, LaserCad, SimuLink, Simscape Multibody, Gazebo, Mujoco
Frameworks	ROS2, Isaac Lab, Holosoma
Programming	C/C++, Python, L ^A T _E X

KEY COURSES UNDERTAKEN

Mechanical Engineering	Thermodynamics, Solid Mechanics, Numerical Analysis, Engineering Mechanics, Analysis and Design of Mechanical Systems, Structural Materials, Mechanical Processing of Material, Fluid Mechanics, Manufacturing Practice Lab
Mathematics & Physics	Multivariable Calculus, Linear Algebra, Differential Equations, Classical Mechanics
Programming	Computer Programming and Utilization, Artificial Intelligence and Data Science, Programming for Data Science

EXTRACURRICULARS

- Participated and completed several **half marathons** with a sub **2 hour 15 min time**. (2025 - Present)
- Bagged the **first prize** in the **Institute Badminton League**, hosted by the Institute Sports Council (2025)
- Secured **first and second position** in the swimming and badminton fresher general championship (2024)
- Successfully completed the **Hampta Pass Trek**(13000 feet+), a **High Altitude Himalayan Trek** (2025)
- Successfully completed a year long course in **Badminton** under the **National Sports Organisation** (2024 - 25)
- Participated and performed in the **Annual Production** conducted by **Fourthwall**, the dramatics club (2025)